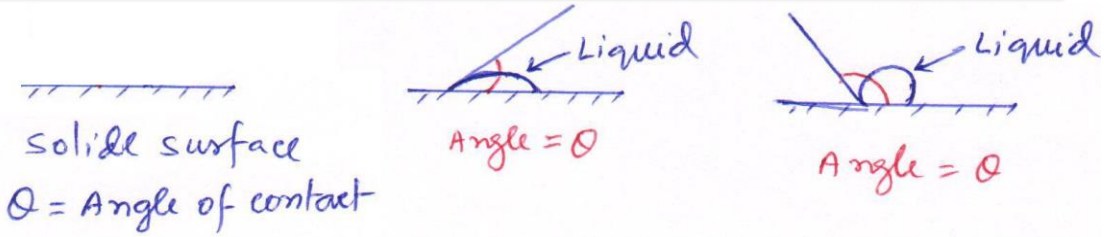
Angle of Contact

Angle of Contact:

The angle between the tangent to the liquid surface at the point of contact and the solid surface, inside the liquid, for that pair of liquid and solid.



i) when $\theta < 90^\circ$, spreading / wetting occurs

e.g. water on glass surface ; $\theta = 18^\circ$

Parafin on glass ; $\theta = 27^\circ$

Oil on clothes, alcohol on glass, etc.

ii) when $\theta > 90^\circ$, no spreading or wetting occurs

e.g. Hg on glass ; $\theta = 110^\circ$

H_2O on solid parafin ; $\theta = 140^\circ$

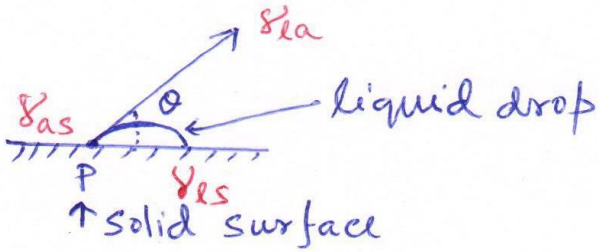
H_2O drop on a lotus surface ; $\theta = 148^\circ$

Superhydrophobicity:-

- ✓ **Superhydrophobic surfaces are highly hydrophobic, i.e., extremely difficult to wet**
- ✓ **The contact angles of a water droplet on a superhydrophobic material exceeds 150°**
- ✓ **Many natural surfaces, such as plant leaf, insects, etc. exist around us**

Expression of Angle of Contact (θ)

Where the two surfaces meet, they form a contact angle, θ, which is the angle the tangent to the surface makes with the solid surface.



γ_{la} = surface tension along liq.-air

γ_{ls} = " " " liq.-solid

γ_{as} = " " " air-solid

If point 'p' is static, then horizontal force at this point (P) must be balanced with each other.

The force right to the point (P) = $\gamma_{ls} + \gamma_{la}\cos\theta$

The force left to the point (P) = γ_{as}

At equilibrium (or under balanced condition)

$$\gamma_{as} = \gamma_{ls} + \gamma_{la}\cos\theta$$

$$\Rightarrow \cos\theta = \frac{\gamma_{as} - \gamma_{ls}}{\gamma_{la}}$$

If $\gamma_{as} > \gamma_{ls} + \gamma_{la}\cos\theta$, then there cannot be any equilibrium and spreading occurs.

If $\gamma_{as} > \gamma_{ls}$, then $\cos\theta > 0 \Rightarrow \theta < 90^\circ$

If $\gamma_{as} < \gamma_{ls}$, then $\cos\theta < 0 \Rightarrow \theta > 90^\circ$

<u>Factors affecting θ</u> :-

(i) Nature of solid surface
(ii) Nature of liquid
(iii) value of interfacial tension
(iv) Temperature of the system.

Spreading Co-efficient

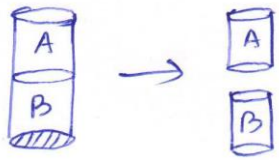
Cohesive force:- **It is the force of attraction between molecules of same kind.**

work of cohesion:

$$\Rightarrow W_{AA} = (\gamma_A + \gamma_A) - 0$$

$$W_{AA} = 2\gamma_A$$

Adhesive force:- **It is the force of attraction between molecules of different kind.**



work of adhesion -

$$\Rightarrow W_{AB} = \gamma_A + \gamma_B - \gamma_{AB}$$

Spreading co-efficient:- **If a small quantity of an immiscible liquid 'A' be placed over another immiscible liquid 'B' then whether 'A' will spread over 'B' or not depends on magnitude of adhesive and cohesive work. The difference between adhesive and cohesive work is known as spreading co-efficient (ϕ).**

Eqn.:

$$\phi = W_{ad} - W_c$$

$$= (\gamma_A + \gamma_B - \gamma_{AB}) - (2\gamma_A)$$

$$\Rightarrow \phi = \gamma_B - \gamma_A - \gamma_{AB}$$

W_{ad} = work of adhesion

W_c = " " cohesion

If ϕ = +ve, i.e., $W_{ad} > W_c$, then A can spread over B.

e.g., oil in water

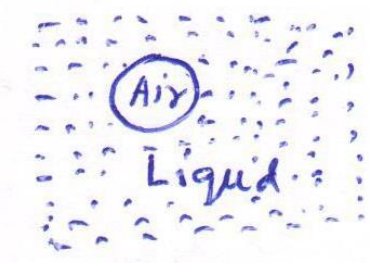
If ϕ = -ve, then $W_{ad} < W_c$; A can not spread over B.

It will form globule.

e.g., Hg on glass

Excess Pressure

(a) Excess pressure inside an air-bubble in a liquid:-



- ✓ **Radius of the bubble = r**
- ✓ **Surface area of the bubble = $4 \pi r^2$**
- ✓ **The excess pressure inside the bubble will try to increase the volume of the bubble as well as the surface area of the bubble outwardly.**
- ✓ **Initial surface energy = $4 \pi r^2 \gamma$**
- ✓ **It can be proved that the excess pressure (P_{ex}) to increase the radius of bubble can be given by;**

$$P_{ex} = \frac{2\gamma}{r}$$

(b) Excess pressure inside a soap bubble:-

- ✓ **A soap bubble forms two liquid surfaces in contact with air, one inside the bubble and the other outside the bubble.**
- ✓ **Thus,**

$$P_{ex} = \frac{4\gamma}{r}$$

What will be the pressure inside an air bubble of 0.2 mm diameter situated just below the surface of water? [γ_{water} = 0.072 N/m]

Temperature Dependence of Surface Tension

Generally, for a liquid: $\gamma \downarrow$ with $T \uparrow$

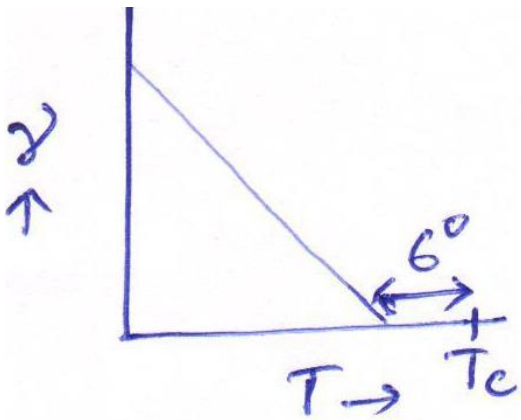
- ✓ **With $T \uparrow$, vaporization $\uparrow$**
- ✓ **Number density in the vapour phase $\uparrow$**
- ✓ **Number density in the liquid phase $\downarrow$**
- ✓ **Thus, the difference in force of attractions between a surface molecule-vapour molecule and surface molecule-liquid molecule in the bulk is diminished**

An emperical relation between γ and temp.:

Eötvös - Ramsay - Shields Equation:

$$(MV)^{2/3}\gamma = K(T_c - T - 6)$$

where, M = Molar mass of liquid
V = specific volume of liquid
T_c = critical temp.
T = Temperature
K = constant



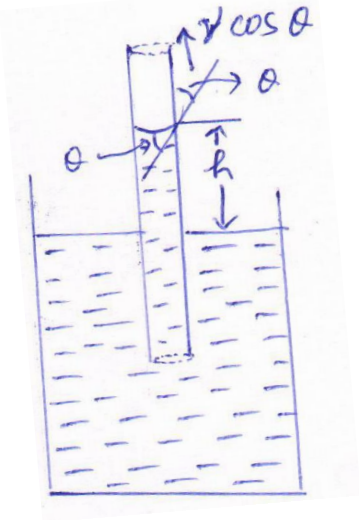
- ✓ **Theoretically, γ = 0 at T_c as no demarcation can be done between a liquid and a vapour phase at and above T_c.**
- ✓ **Thus, no surface tension exists at T_c**
- ✓ **Experimentally, γ = 0 at a temperature below 6° of T_c**
- ✓ **Indicated in Eötvös-Ramsay-Shields relation**

Measurement of Surface Tension

Methods:-

(1) Capillary rise method
(2) Drop-wet method
(3) Drop-counting method
(4) The ring detachment method

(1) Capillary rise method:-



r = Radius of capillary
θ = Angle of contact
γ = surface tension of liquid
ρ = Density of the liquid
h = Height of the liquid column inside capillary.

Total upward force $= 2\pi r \gamma \cos\theta$
Total downward force $= \left(\pi r^2 h + \frac{1}{3}\pi r^3\right)\rho g$

At equilibrium, $2\pi r \gamma \cos\theta = \left(\pi r^2 h + \frac{1}{3}\pi r^3\right)\rho g$

$$\Rightarrow \gamma = \frac{\left(rh + \frac{1}{3}r^2\right)\rho g}{2\cos\theta}$$

Often θ is found to be nearly zero, so $\cos\theta = 1$
Also, r is small, thus neglecting r^2, $\gamma \simeq \frac{rh\rho g}{2}$

Thus, the measurement of *h* and *r* with a knowledge of the liquid density at a particular temperature will enable us to determine the surface tension.

Viscosity of Gases

- ✓ **Viscosity of gases**
- ✓ **Definition of Viscosity Co-efficient (η)**
- ✓ **Relation between viscosity co-efficient and mean free path (l):** $\eta = \frac{1}{3} \rho \bar{c} l$
- ✓ **Effect of temperature on the viscosity of gases:**

The expression of viscosity of gases from the kinetic model shows that viscosity of a gas increases with increase of temp.,

$$\eta = \frac{2}{3} \frac{\sqrt{MRT}}{\pi^{3/2} \sigma^2 N_A}$$

$$\eta \propto \sqrt{T}$$

Viscosity

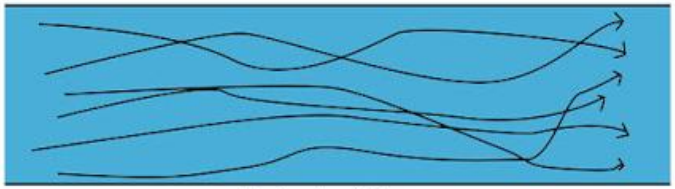
Streamline or Laminar Flow:-

Streamline Flow

- ✓ Velocity at each and every point remains almost the same in magnitude and direction
- ✓ The path followed by a particle in streamline flow is called streamline
- ✓ No two streamline would intersect each other
- ✓ Reynolds' number ranges from 0-2000

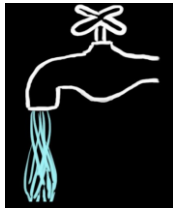
Example: Flow of a fluid through a capillary tube

Turbulent Flow:-



Turbulent Flow

- ✓ Velocity at different point may vary with time in direction and magnitude
- ✓ There are rotatory motion involved within the flow
- ✓ The path of the flow is known as turbulent
- ✓ Two turbulent line can cross each other
- ✓ Reynolds' number is above 2000



Example: Flow of a fluid through water pipe

Reynold's Number (R):-

- ✓ Reynold number is a number to characterize a flow as steady or turbulent. It depends on;
  - (i) Viscosity co-efficient of liquid (η)
  - (ii) Density of the liquid (ρ)
  - (iii) Diameter of the tube through which the liquid is flowing (d)

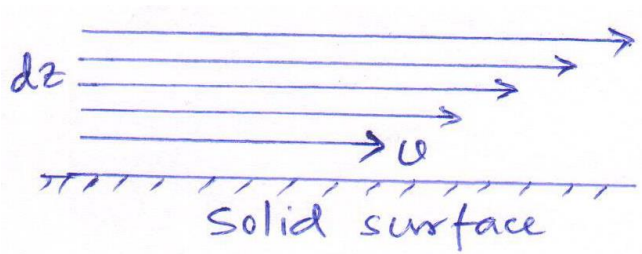
$$R = \frac{d \times u \times \rho}{\eta}$$; u: Average velocity of flow.

$R < 2000 \Rightarrow$ Streamline flow.

$R > 2000 \Rightarrow$ Turbulent flow.

Viscosity

Cause of Viscosity:-



Newtonian Fluid:-

- ✓ **If flow of a fluid is such that the tangential force or viscous force or frictional force per unit area is proportional to velocity gradient then it is known as Newtonian fluid.**
- ✓ **Examples: Flow of any homogeneous fluid such as water, alcohol, etc.**

$$f = -\eta A \frac{dv}{dz}$$

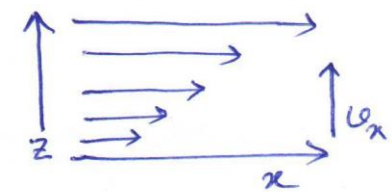
; f = Tangential backward viscous force

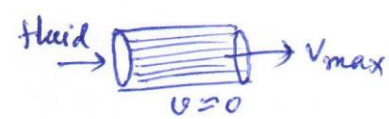
A = Area of contact of two liquid layers

dz = distance betwn. two layers

dv = velocity difference of two adjacent layers

$\frac{dv}{dz}$ = velocity gradient [Rate of change of velocity w.r.t. distance from the wall]





$$\therefore \frac{f}{A} \propto \frac{dv}{dz}$$

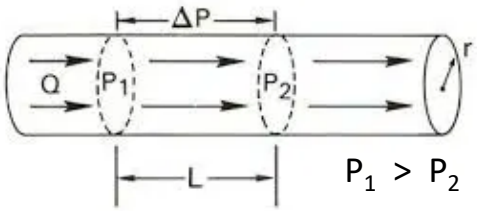
($\frac{f}{A}$ is also called shearing stress)

-ve sign here indicates that 'f' is opposite to the direction of the flow of liquid.

Poiseuille's Equation

Assumptions:-

- ✓ **Fluid is Newtonian, i.e., shearing stress (tangential force per unit area) is proportional to the velocity gradient.**
- ✓ **The liquid layer in contact with the tube wall is considered to be at rest.**
- ✓ **Pressure difference at the two end of the tube is maintained so that the fluid flows with a uniform velocity.**
- ✓ **Pressure is same at all points along the radius, i.e., there is no radial flow.**
- ✓ **Motion is streamline and parallel to the axis of the tube.**



$P_1 > P_2$

Poiseuille's Equation:

$$\eta = \frac{\pi (P_1 - P_2) r^4}{8 l \frac{dV}{dt}}$$

Fluidity (ϕ):-

Fluidity (ϕ) is the reciprocal of viscosity co-efficient (η) of a liquid: $\phi = \frac{1}{\eta}$

Hence, it is a measure of flowing capacity of a liquid. Fluidity depends on several factors;

(1) Liquid having heavier and longer molecules flow less easily than lighter and smaller molecules.

(2) Liquids with spherical molecules flow more easily than that having planar molecules.

(3) Presence of any impurity changes fluidity.

Temperature Dependence of η of a Liquid

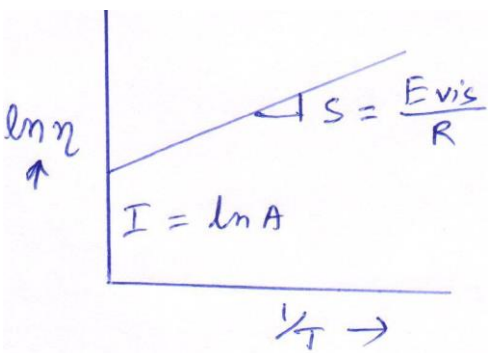
- ✓ **When a molecule in a liquid flows, it has to acquire sufficient energy to squeeze past other molecules overcoming the force of intermolecular attraction.**
- ✓ **This minimum energy is known as viscosity activation energy.**
- ✓ **The number of molecules having this energy or greater than that is given by Boltzmann factor;**

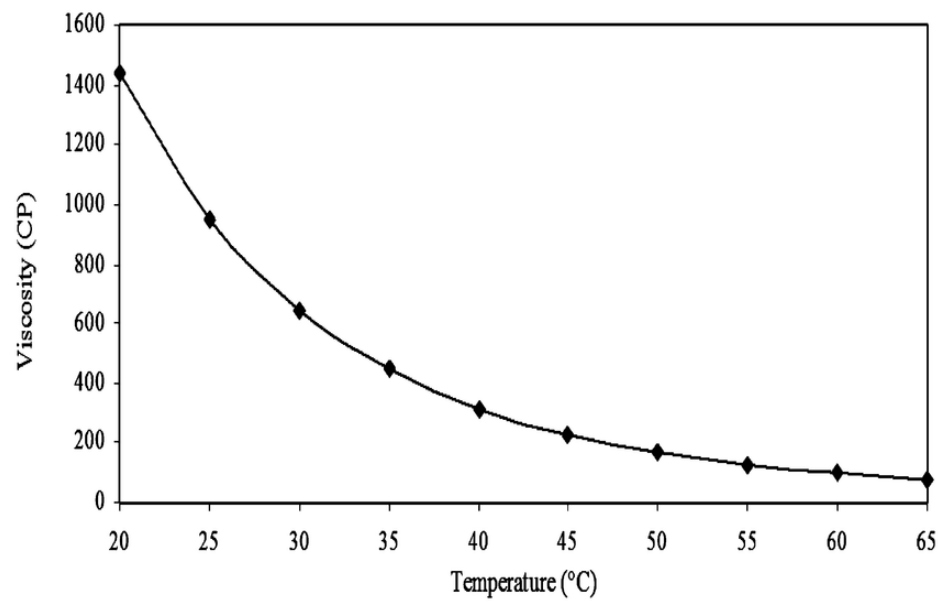
$$\varphi \propto e^{-E_{vis}/RT}$$

$$\Rightarrow \eta \propto e^{E_{vis}/RT} \quad \left(\because \varphi = \frac{1}{\eta}\right)$$

$$\Rightarrow \eta = A e^{E_{vis}/RT} \quad [A = \text{Proportionality constant}]$$

$$\ln \eta = \ln A + \frac{E_{vis}}{RT}$$





- ✓ **η of liquids decrease with rise in temperature**

Experimental Determination of η:-

(1) Using falling sphere viscometer
(2) By Ostwald viscometer
(3) Rotating cylinder method
etc.

Various Types of Viscosities

- ✓ Viscometry
- ✓ Not an absolute method

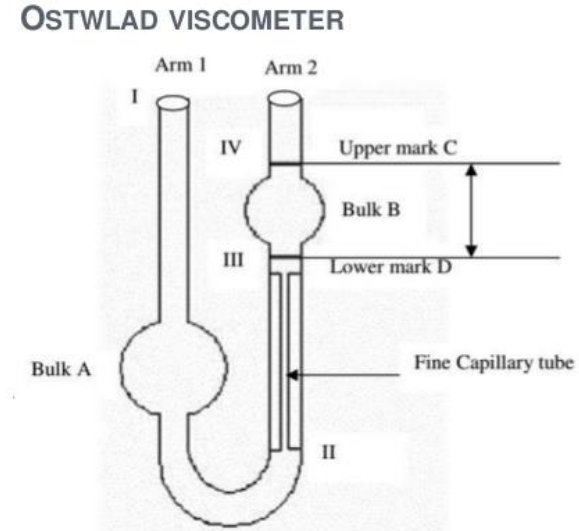


Table: Dilute Solution Viscosity Designations

Common Name	IUPAC *Name*	Definition
Relative viscosity	Viscosity ratio	$\eta_{rel} = \frac{\eta}{\eta_o} = \frac{t}{t_o}$
Specific viscosity	-	$\eta_{sp} = \frac{\eta - \eta_o}{\eta_o} = \frac{t - t_o}{t_o} = \eta_{rel} - 1$
Reduced viscosity	Viscosity number	$\eta_{red} = \frac{\eta_{sp}}{C} = \frac{\eta_{rel} - 1}{C}$
Inherent viscosity	Logarithmic viscosity number	$\eta_{inh} = \frac{\ln \eta_{rel}}{C}$
Intrinsic viscosity	Limiting viscosity number	$[\eta] = \left(\frac{\eta_{sp}}{C}\right)_{C=0} = (\eta_{inh})C = 0$

η_0 = viscosity of pure solvent
η = " " solution

C = conc.

$\equiv \lim_{C \to 0} \frac{\eta_{sp}}{C} \equiv \lim_{C \to 0} C^{-1} \ln \eta_{red}$

Inverse of conc. dL/g
conc. unit g/dL (grams per 100 mL)

The extrapolated value of $\frac{\eta_{sp}}{C}$ at zero conc.